

Real-Time Vessel Monitoring Using Low Cost RTL-SDR with Web-Based AIS Visualization

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Abstract— Ship traffic monitoring plays a vital role in observation, research, maritime security, and navigation safety. The Automatic Identification System (AIS) is an international standard for vessel tracking, creating a demand for efficient and affordable AIS receiver solutions. This paper presents a low-cost AIS receiver system built with RTL-SDR and Raspberry Pi devices. The system uses a C-based decoder to perform GMSK demodulation, NRZI decoding, and CRC16 error checking, ensuring high data accuracy. Decoded information is stored in a MariaDB database and visualized in real time through a web-based interactive map. The platform displays vessel positions, movements, and detailed information, with additional features such as ship filtering, playback controls, weather overlays, and distance measurement tools. Field tests demonstrate that the system achieves over 99% decoding accuracy with minimal latency, making it suitable for localized coastal monitoring. This approach offers a scalable and cost-effective alternative to conventional AIS receivers, supporting both research and operational applications in diverse maritime environments.

Keywords— Automatic Identification System (AIS), RTL-SDR, Raspberry Pi, MariaDB, Web-based Visualization.

I. INTRODUCTION

Maritime traffic monitoring is essential to ensure safe navigation and prevent potential maritime accidents, particularly in areas with high vessel density such as port regions [1]. An effective monitoring system plays a critical role in supporting these safety efforts [2].

The Automatic Identification System (AIS) is a standardized technology widely adopted in the maritime industry for real-time vessel tracking [3] [4]. AIS allows ships to transmit key information such as position, identity, speed, and course to nearby vessels and coastal stations [5]. This data is valuable for monitoring maritime traffic, detecting navigation route violations, and enhancing search and rescue (SAR) operations [6].

Although AIS receivers play an important role, conventional devices are often expensive and require substantial computing resources [7]. This creates challenges for small-scale monitoring stations and research environments seeking to implement maritime monitoring systems on limited budgets.

To address these challenges, this study proposes an AIS receiver system based on RTL-SDR and Raspberry Pi, offering an efficient, low-cost, and easily deployable solution [8]. The system employs decoding algorithms implemented in the C programming language, applying GMSK demodulation,

NRZI decoding, and CRC16 error checking to improve data accuracy.

The automatically decoded data is stored in a MariaDB database and visualized through a web interface using Leaflet, enabling real-time monitoring of vessel positions on an interactive map. This system is intended to provide a practical solution for small-scale maritime monitoring applications, particularly in research environments or areas with limited resources.

The main contribution of this paper lies in the development of a lightweight and low-cost AIS receiver system using RTL-SDR and Raspberry Pi, combined with an efficient AIS signal decoding algorithm implemented in the C programming language, optimized for devices with limited resources. The system integrates real-time data storage using MariaDB and visualizes vessel positions on a web-based interactive map powered by Leaflet.js. Additionally, the system's performance is validated through both laboratory and field testing in real maritime environments, including high-density vessel areas such as Suramadu and Surabaya North Quay.

Furthermore, this paper is organized as follows: Chapter II reviews related work, Chapter III describes the system design, Chapter IV presents the implementation and results, and Chapter V concludes the study.

II. RELATED WORK

In greater detail, this section discusses several studies conducted by previous researchers on the development of AIS systems.

Oumimoun et al. [9] studied the reception and decoding of AIS signals from Class A moving stations using RTL-SDR devices equipped with VHF antennas, utilizing GNU Radio for AIS signal processing and decoding.

Alba et al. [10] designed a local monitoring application for the Automatic Identification System (AIS) on ships using Android Studio with the Java programming language, integrating the Google Maps API for real-time vessel movement visualization and Firebase as the database.

Putra et al. [11] developed an AIS-based ship monitoring and identification system using Laravel and Vue.js, applying the Agile Development methodology to support a rapid and flexible development process.

Building upon previous research [9]–[11], this study focuses on improving AIS signal reception and vessel

monitoring through a low-cost, real-time system. The primary distinctions between prior works and the proposed system are summarized in Table 1.

TABLE I. COMPARISON WITH PREVIOUS STUDIES

Previous Studies	Object	Method
[9]	AIS signal reception and decoding	VHF antenna + RTL-SDR + GNU Radio
[10]	Local AIS monitoring application	Android Studio + Google Maps API + Firebase
[11]	Ship monitoring and identification	Laravel + Vue.js + Agile Development
Proposed System	AIS signal reception and monitoring	VHF antenna + RTL-SDR + Raspberry Pi + C-based AIS decoding + MariaDB + Leaflet

III. SYSTEM DESIGN

Figure 1 presents the proposed system architecture for detecting and visualizing the real-time positions of vessels using an AIS receiver based on RTL-SDR and Raspberry Pi. The system operates as a local monitoring solution, focusing on vessel traffic within the coverage area of a VHF antenna. The antenna is connected to an RTL-SDR device that captures AIS signals, which are subsequently processed by a Raspberry Pi running a custom AIS decoder implemented in C. The decoded data are stored in a MariaDB database and visualized on an interactive map using Leaflet, enabling intuitive and real-time vessel monitoring. At this stage, the system is designed as a standalone single-node solution operating on localhost, and does not yet support data aggregation from multiple receivers or remote stations.

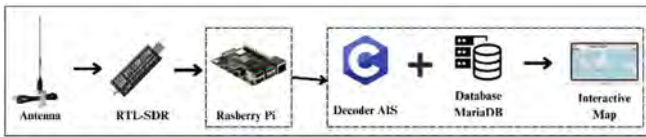


Fig. 1. System Design

To provide a structured overview, this study follows the process illustrated in the flowchart shown in Figure 2. Data received by the Raspberry Pi from the VHF antenna and RTL-SDR is processed and stored in a database, then displayed on a website interface. The system operates in real time, thereby improving the efficiency and effectiveness of ship monitoring.

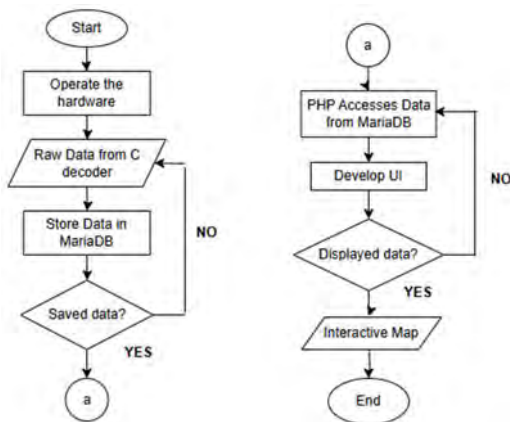


Fig. 2. Flowchart System

This approach was selected due to the suitability of implementing the decoder in the C programming language on the Raspberry Pi, which is optimized for devices with limited

computational resources. Compared to more complex frameworks like GNU Radio, C enables lower resource consumption and improved performance [9]. The Raspberry Pi was chosen as the core processing unit for its compact size, affordability, and adequate capability to decode AIS data and manage database operations. The RTL-SDR device was selected as the AIS receiver for its low cost [12], flexibility, and ability to receive VHF signals at the standard AIS frequencies of 161.975 MHz and 162.025 MHz. MariaDB was used as the database management system for its lightweight nature, reliable performance, and compatibility with the Raspberry Pi, ensuring efficient data storage and retrieval.

Leaflet.js is employed for the interactive map interface due to its lightweight design, ease of customization, and efficient rendering capabilities, making it well-suited for real-time vessel position visualization with minimal resource usage. The integration of this hardware, software, and framework combination is expected to ensure stable performance and support real-time operations on resource-constrained devices.

IV. IMPLEMENTATION AND RESULT

A. Experimental Setup and Environment

The experimental setup was implemented using a Raspberry Pi 3 Model B+ with 1 GB of RAM as the primary processing unit. An RTL-SDR V4 dongle was used for AIS signal reception, paired with a VHF antenna tuned to the standard AIS frequencies of 161.975 MHz and 162.025 MHz. The Raspberry Pi operated on Raspberry Pi OS (32-bit) with a 32 GB memory card. The AIS decoding application was developed in the C programming language to optimize performance on a resource-limited platform. RTL-SDR was selected for its affordability and proven effectiveness in AIS signal acquisition [12], making it ideal for real-time data collection in a compact system. All configurations including RTL-SDR frequency and gain settings, MariaDB database management, and Leaflet.js-based visualization were executed directly on the Raspberry Pi. The hardware setup is illustrated in Figure 3.

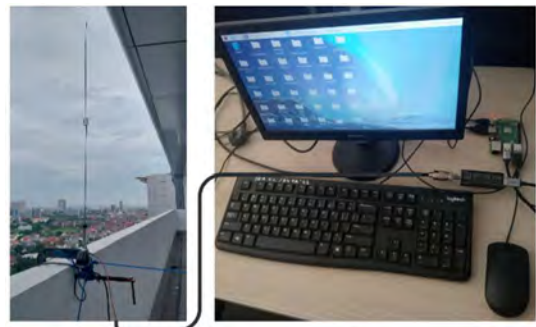


Fig. 3. Equipment Setup

The RTL-SDR device is configured to a center frequency of 162 MHz with a sampling rate of 300 kHz, allowing it to capture AIS signals at 161.975 MHz and 162.025 MHz, which fall within the effective bandwidth. The gain is set to 80 dB, as this value yielded the most stable and responsive data reception during testing. Data streaming is established via TCP/IP communication using a local IP address and port 2345.

The VHF antenna was installed on the 12th floor of the PENS campus building in Surabaya, providing an elevated position capable of receiving AIS signals from vessels located

up to approximately 40 km away. The testing environment included real-world conditions such as environmental noise and high vessel traffic density, offering a practical scenario for evaluating data reception performance.

MariaDB is employed as the database system due to its lightweight architecture, efficient data handling, and seamless integration with PHP for data access. The web interface is developed using JavaScript, HTML, and CSS, which interact with PHP to retrieve and display data on interactive maps rendered using Leaflet.js. Leaflet.js is selected for its efficient rendering capabilities and low resource consumption, enabling smooth real-time visualization of vessel positions.

Additional field testing was also conducted at two alternative locations—Suramadu Bridge and Surabaya North Quay—to further evaluate the system’s reception performance in environments with high maritime traffic density. Suramadu Bridge spans across the Madura Strait and serves as a key corridor for vessels entering and exiting the Port of Surabaya, while Surabaya North Quay is situated near the Tanjung Perak Port, one of Indonesia’s busiest seaports. These locations were chosen for their proximity to dense shipping routes and minimal physical obstructions. Detailed results from these outdoor tests are discussed in the Testing Environment section.

B. Signal Processing and AIS Data Visualization

After setting up the experimental environment, IQ samples are acquired from the RTL-SDR and transmitted to the Raspberry Pi via TCP/IP for processing. The IQ signal undergoes several stages: first, noise and out-of-band signals are filtered to isolate relevant AIS signals. The filtered signal is then demodulated using the Gaussian Minimum Shift Keying (GMSK) method, converting the analog signal to a digital bit stream. The bit stream is processed through bitwise operations to conform to the AIS protocol structure. Decoding is then performed using the High-Level Data Link Control (HDLC) protocol to extract the AIS payload. To ensure data integrity, a CRC16 (Cyclic Redundancy Check) method is applied, allowing only valid data to be processed further. The verified payload is parsed to extract critical information, including the Maritime Mobile Service Identity (MMSI), geographical position (longitude and latitude), speed over ground (SOG), course over ground (COG), and timestamp. This parsed data is displayed in real-time on the Raspberry Pi terminal, as shown in Figure 4.

```
port: 2345; alamat: 10.252.135.227
```

=== (12 bytes) RTL0 ===

	MID	MMSI	longitude	latitude	speed	course	waktu
1	477871800	112.665748	-7.038337	27 km/h	19.4	15_36_5	
3	567421000	112.676467	-7.175233	0 km/h	38.0	15_36_5	
1	525300005	112.724980	-7.202723	0 km/h	197.4	15_36_6	
1	525001047	112.729847	-7.207408	0 km/h	152.2	15_36_7	
1	525101059	112.729778	-7.207437	0 km/h	292.8	15_36_8	

Fig. 4. Raw Data In Terminal

id	MID	MMSI	Longitude	Latitude	Speed	Course	Waktu
1501	1	416037000	112.703245	-7.190382	9	122.2	143157
1502	1	525501876	112.730737	-7.196765	0	267.6	143159
1503	1	525100588	112.722885	-7.203673	0	42.6	14326
1504	1	525501876	112.730735	-7.196765	0	267.6	14328

Fig. 5. Database

After decoding, vessel information is parsed and stored in a MariaDB database. The data includes MID, MMSI, geographical position, vessel speed, course, and timestamp, as shown in Figure 5. Data processing ensures the information adheres to the required format before being entered into the database.

The stored data is then visualized in real-time on a map using the Leaflet.js platform, as shown in Figure 6. In addition to plotting vessel positions, the interactive map offers multiple user-friendly features to enhance usability and data exploration. These include a navbar for navigation, a search bar for querying ships by MMSI, zoom controls, and map theme toggles (Base, Light, Dark, and Satellite). Users can interact with vessel markers to reveal a pop-up containing detailed ship information such as MMSI, speed, course, last update time, vessel type, and country of origin. To improve clarity and visual differentiation, vessel markers are color-coded based on their ship types. A hover effect on the navbar enhances usability by displaying descriptive tooltips when icons are hovered over.

Furthermore, the system provides advanced filtering capabilities, allowing users to filter vessels by type, status (anchored/moving), and country of origin. A weather overlay feature is also integrated, displaying wind, temperature, and precipitation layers, each with toggles and color scales for intuitive understanding. A distance measurement tool enables users to calculate the distance between two points directly on the map.

For deeper insight, a vessel history tracking feature enables playback of a ship's movement path based on MMSI data, complete with playback controls (play, pause, speed up to 10x) and a chronological timeline panel. Finally, the map includes a statistical dashboard that visualizes ship distribution by type, status, origin country, and top ship owners using interactive charts and bar graphs. A data table at the bottom displays recent vessel activity logs.

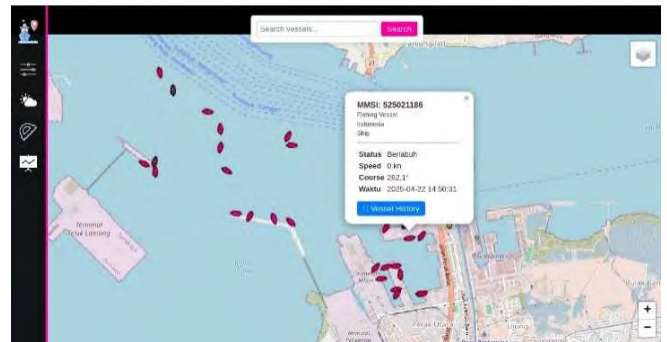


Fig. 6. Website Visualization

C. Performance Evaluation

The performance evaluation of the AIS receiver system is based on several key parameters. First, the signal reception level is assessed by counting the number of AIS messages successfully received within a 2-minute observation period. The system consistently received messages at an average interval of 15 seconds across both AIS frequencies (161.975 MHz and 162.025 MHz). Additionally, decoding accuracy is evaluated using a C language-based algorithm, which includes GMSK demodulation, NRZI decoding, and CRC16 error checking. Decoding accuracy is determined by

comparing the number of successfully decoded messages to the total messages received, as expressed in Equation (1).

$$\text{Decoding Accuracy (\%)} = \frac{\text{Correctly Decoded Messages}}{\text{Messages Received}} \times 100\% \quad (1)$$

The results for signal reception levels and decoding accuracy during the observation period are presented in Table 2. Beyond signal-level performance, it is also essential to evaluate the responsiveness and efficiency of the entire system pipeline.

TABLE II. SIGNAL RECEPTION AND DECODING ACCURACY OVER 2-MINUTE OBSERVATION PERIOD

Interval Time	Messages Received	Correctly Decoded Messages	Decoding Accuracy (%)
00:00 – 00:15	310	308	99.35%
00:15 – 00:30	320	320	100%
00:30 – 00:45	315	312	99.05%
00:45 – 01:00	325	322	99.08%
01:00 – 01:15	300	298	99.33%
01:15 – 01:30	305	303	99.34%
01:30 – 01:45	310	310	100%
01:45 – 02:00	320	318	99.38%

To evaluate the system's performance in processing and displaying AIS data in real-time, time measurements are taken at three key stages: decoding AIS data, storing the data in the MariaDB database, and visualizing the data on a web-based map. The results of the time measurements for each stage are shown in Table 3.

TABLE III. TIMING MEASUREMENTS FOR AIS DATA DECODING, DATABASE INSERTION, AND VISUALIZATION

Message ID	Decoding Time (hh:mm:ss)	Insert Time (hh:mm:ss)	Visualization Time (hh:mm:ss)
101	10:12:35	10:12:35	10:12:36
102	10:12:35	10:12:35	10:12:36
103	10:12:35	10:12:36	10:12:38
104	10:12:35	10:12:36	10:12:38
105	10:12:35	10:12:36	10:12:38
106	10:12:36	10:12:36	10:12:38
107	10:12:36	10:12:36	10:12:38
108	10:12:36	10:12:37	10:12:39
109	10:12:36	10:12:37	10:12:39
110	10:12:36	10:12:37	10:12:39

D. Testing Environment

To assess the influence of geographic location, antenna placement, and environmental conditions on AIS signal reception, additional testing was conducted at three distinct locations in Surabaya: the PENS rooftop (12th floor), Suramadu Bridge, and Surabaya North Quay. The objective was to compare the quantity and quality of AIS data received under varying levels of maritime activity, RF signal

conditions, and line-of-sight obstructions using a consistent setup consisting of an RTL-SDR dongle, Raspberry Pi, and a VHF antenna. The tests were performed at two different times—15:20 and 18:31 WIB—using the same RTL-SDR dongle, Raspberry Pi, and a VHF antenna setup to ensure consistency. Each session lasted 20 minutes and was divided into four 5-minute intervals. Tables IV and V present the number of AIS messages received at each location during the respective timeframes.

TABLE IV. AIS DATA RECEIVED AT 15:20

Interval Time	Number of Data Received at PENS	Number of Data Received at Suramadu
15:20 – 15:25	43	200
15:26 – 15:30	121	395
15:31 – 15:35	201	600
15:36 – 15:40	269	909

TABLE V. AIS DATA RECEIVED AT 18:31

Interval Time	Number of Data Received at PENS	Number of Data Received at Surabaya North Quay
18:31 – 18:36	81	1.928
18:37 – 18:41	143	2.368
18:42 – 18:46	171	4.954
18:47 – 18:51	208	5.001

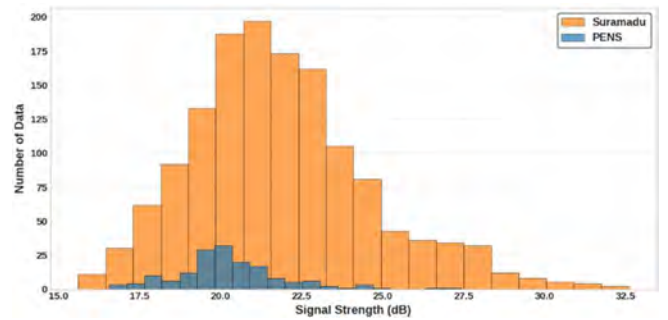


Fig. 7. Comparison between PENS and Suramadu in terms of AIS signal strength distribution.

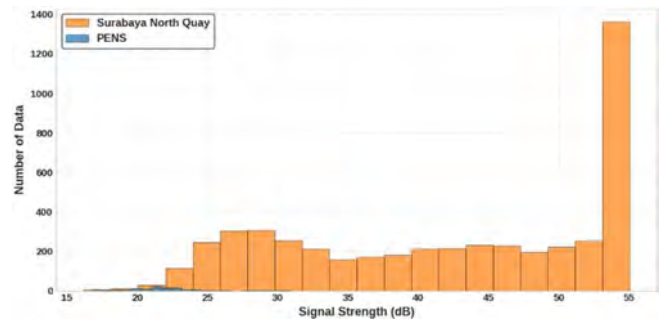


Fig. 8. Comparison between PENS and Surabaya North Quay in terms of AIS signal strength distribution.

Based on Table IV and Table V, during the first session at 15:20, the PENS campus received 269 AIS messages, while Suramadu recorded 909. In the second session at 18:31, PENS received 208 messages, whereas Surabaya North Quay recorded 5,001. Although the antenna at PENS was installed on the 12th floor, which theoretically provides a height advantage for extending signal reception range, the results show that Suramadu and Surabaya North Quay still achieved significantly higher reception performance. This difference can be clearly seen in the signal strength charts above, where

Suramadu and North Quay display stronger and more consistent signal levels compared to PENS.

Figures 7 and 8 further illustrate the comparative signal strength distributions across the observation sites. In Figure 7, the histogram shows that Suramadu consistently recorded stronger signal levels than PENS. Suramadu's signal strength is centered around 21–24 dB, with a smooth distribution and higher data volume across nearly the entire range, peaking at close to 200 messages in the most frequent bin. In contrast, PENS exhibited a narrower distribution concentrated around 18–21 dB, with significantly fewer messages per bin. This suggests that despite the height advantage at PENS, its reception is constrained by environmental limitations and lower vessel density in its surroundings.

In Figure 8, the contrast is even more pronounced. The signal strength at Surabaya North Quay spans a broader and higher range, with a prominent peak at 54 dB and over 1,300 messages in that bin alone. This indicates exceptional reception quality, likely due to the location's proximity to major shipping lanes and unobstructed line-of-sight over the sea. Conversely, PENS remains confined to lower signal strength bins with a relatively flat distribution and minimal data count, reaffirming its weaker performance. These histograms reinforce the conclusion that both Suramadu and Surabaya North Quay offer more favorable conditions for AIS signal reception compared to the elevated PENS rooftop.

TABLE VI. COMPARISON OF THE VISUAL APPEARANCE OF THE WEBSITE AT 15:20

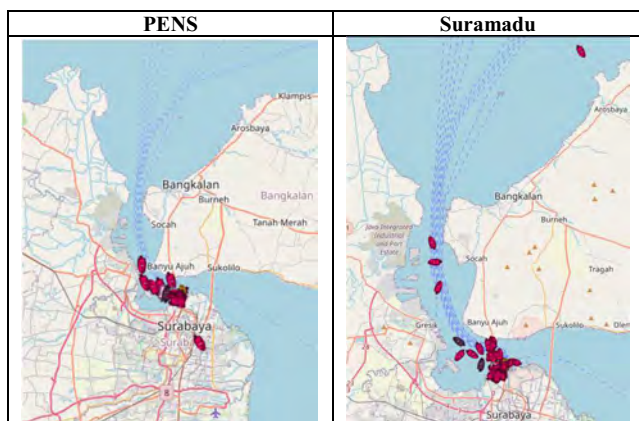
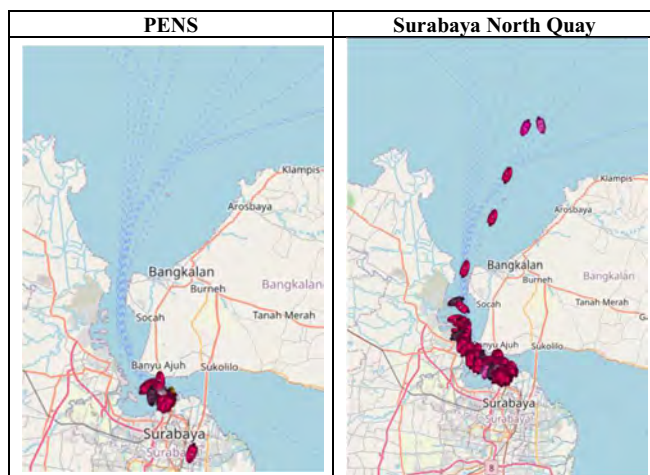


TABLE VII. COMPARISON OF THE VISUAL APPEARANCE OF THE WEBSITE AT 18:31



Map visualizations generated by the web-based AIS system further confirm the disparities in reception quality among the three locations. At 15:20, the web interface for PENS displayed only a limited number of static vessel icons with minimal movement, indicating sparse data reception and restricted coverage. In contrast, the Suramadu interface showed a significantly higher number of vessel markers, including dynamic icons representing real-time position updates and broader maritime coverage. Similarly, at 18:31, the Surabaya North Quay interface revealed dense maritime traffic, with vessel icons traveling in various directions—signaling high data throughput and excellent signal clarity. Tables VI and VII present a side-by-side visual comparison of the web-based output for each location during the relevant observation window.

Moreover, the number of unique vessels successfully received at PENS was considerably lower, with a detection range limited to approximately 12–16 km. Fast-moving vessels were also rarely detected from PENS, whereas both Suramadu and Surabaya North Quay registered numerous vessels traveling at speeds of up to 20–27 knots, with signal detection ranges exceeding 22 km.

These visual and quantitative results collectively highlight the strong dependence of AIS system performance on geographic positioning and environmental context. Factors such as physical obstructions, line-of-sight availability, antenna elevation, and proximity to high-traffic maritime zones are pivotal. Furthermore, the effectiveness of the low-cost receiver system—when deployed under optimal conditions—demonstrates its potential for scalable and practical use in real-world coastal monitoring applications. This insight is valuable for guiding strategic placement of AIS ground stations, particularly in archipelagic and urban port regions.

E. Comparative Analysis

Based on the evaluation of signal reception levels, data decoding accuracy, database update efficiency, and web visualization latency, the proposed system demonstrates significant advantages compared to previous implementations.

Compared to the study [9], which employed GNU Radio for decoding AIS signals, the proposed system achieves equivalent or better reception rates and decoding accuracy, but with significantly lower processing power consumption. This is accomplished by utilizing a lighter, optimized C-based decoder for Raspberry Pi, which allows for more efficient real-time processing and reduces the complexity commonly found in GNU Radio. Consequently, this system can operate more responsively and with reduced power consumption, making it suitable for applications on devices with limited resources.

Compared to the study [10], which utilized a cloud-based AIS monitoring application with Firebase, the proposed system demonstrates higher efficiency in database updates, as it avoids cloud-related latency and ensures faster data storage and retrieval by using MariaDB locally.

Compared to Study [11], which focuses on the development of a web-based AIS monitoring application using the Laravel and Vue.js frameworks, the proposed system offers advantages in terms of real-time processing and lower web visualization latency. This system directly

processes AIS signals and displays maps instantaneously, without reliance on more complex web frameworks that require additional resources. By utilizing pure HTML, CSS, and JavaScript for the user interface and Leaflet.js for map visualization, this system is lighter and more efficient, enabling rapid and responsive map updates while reducing the latency typically associated with larger web frameworks such as Laravel and Vue.js.

Overall, the system demonstrated stable signal reception, with a decoding accuracy exceeding 99%. The database update time was approximately 0-1 seconds, while web visualization latency was around 1-2 seconds, all while maintaining low resource consumption. These results indicate that real-time vessel monitoring can be effectively achieved even with minimal hardware in resource-constrained environments.

V. CONCLUSION

This paper presented the development and evaluation of a real-time AIS receiver system as a cost-effective solution for localized vessel monitoring. The system achieved stable signal reception, decoding accuracy above 99%, and minimal latency in both data storage and visualization, ranging from 0–2 seconds, while maintaining low power consumption. By utilizing a lightweight C-based decoder on a Raspberry Pi, combined with MariaDB for local storage and Leaflet.js for web-based visualization, the system efficiently processed and displayed AIS data in real time. Compared to previous studies, it offered comparable performance with significantly lower resource requirements, making it suitable for use in constrained environments. Field testing results revealed that proximity to active shipping lanes such as at Suramadu and especially to major port areas like Surabaya North Quay significantly increased the volume of AIS data received, outperforming elevated but obstructed locations such as the 12th floor of the PENS campus. While the current implementation focuses on single-node deployment, future work may expand the system to support multi-node configurations, improve resilience to noisy signals, and enable scalable applications for broader coastal monitoring.

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